

Mark A. Kheyfets

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mkheyfets-tech.github.io/robotics-portfolio

EDUCATION

Worcester Polytechnic Institute
B.S., Robotics Engineering

GPA: 3.6 / 4.0

Worcester, MA
Expected May 2027

Relevant Coursework: Statics, Stress Analysis, Calculus I–IV, Differential Equations, Linear Algebra, Physics

TECHNICAL SKILLS

- **CAD & Drawings:** SolidWorks (Advanced Parts, Assemblies, Drawings, Exploded Views, Sheet Metal Basics), Fusion 360, Geometric Dimensioning & Tolerancing, Bill of Materials (BOM) Management
- **Analysis & Simulation:** SolidWorks FEA (Stress & Deflection), Static & Dynamic MATLAB Motion Simulation, Free-Body Diagrams (FBDs)
- **Fabrication & Prototyping:** 3D printing (FDM/PLA), laser cutting, material selection
- **Sensors & Testing:** Strain-Gauge Load Cells, Wheatstone Bridges, Instrumentation Amplifiers (AD620), ADCs (HX711), Sensor Calibration, Noise Filtering
- **Programming & Tools:** MATLAB, C/C++, Python, GitHub, MS Office

ENGINEERING PROJECTS

2-DOF Parallel Manipulator — Design, FEA, and Fabrication

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Custom load-bearing 5-bar linkage arm and gripper for a robot that pulls boxes off multi-level shelves.

- Designed a load-bearing 5-bar linkage mechanism, custom end-effector, and structural mounting hardware in SolidWorks, optimizing geometry to minimize weight while maximizing frame rigidity.
- Created detailed engineering drawing packages and maintained a complete Bill of Materials (BOM), specifying standard fasteners, bearings, standoffs, and motor drive hardware.
- Conducted SolidWorks FEA stress and deflection analyses on laser-cut acrylic links and 3D-printed load-bearing gears, locating stress risers to ensure a $>2.0\times$ Factor of Safety under payload.
- Formulated free-body diagrams (FBDs) and static/dynamic stability calculations in MATLAB to define safe payload capacity and prevent chassis tipping at maximum arm reach.
- Executed motor and gear sizing calculations, specifying a 20:1 encoder-gear DC motor and high-torque servo to satisfy worst-case stall torque requirements.
- Fabricated physical prototypes using laser cutting and FDM 3D printing, performing fit-up testing and modifying CAD geometries for tighter manufacturing tolerances.

Instrumented Mobile Robot — Custom Load Cell and Sensor Work

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Drive robot with a scale built from scratch and a stack of cleaned-up analog sensors.

- Engineered an integrated digital weighing scale using a strain-gauge load cell and Wheatstone bridge circuit, conditioning signals with an AD620 instrumentation amplifier and HX711 ADC.
- Calibrated sensor arrays against known reference weights using linear regression in MATLAB, implementing rolling-average noise filtering to achieve high-precision repeatability.
- Eliminated ultrasonic and infrared signal jitter by integrating a hardware Schmitt trigger and low-pass software filters, preventing control loop oscillation during close-range navigation.
- Implemented sensor fusion by combining wheel encoder quadrature counts with IMU telemetry, tuning PI control loops to ensure precise straight-line trajectory tracking and repeatable 90-degree turns.

3-DOF Robotic Arm — Motion Math and Camera Calibration

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Sorting arm that finds parts with an overhead camera and moves smoothly to pick them up.

- Developed forward and inverse kinematic equations in MATLAB, generating optimized motion profiles that strictly maintained joint velocity and acceleration limits.
- Calibrated an overhead vision system against a 25 mm reference grid, executing lens distortion correction and coordinate transformations to map pixel space into absolute 3D Cartesian coordinates.
- Systematically debugged spatial error sources across vision calibration, sensor feedback, and mechanical backlash to achieve high end-effector repeatability across sequential pick-and-place cycles.

EXPERIENCE

Research Analyst, OpenStar Technologies

Spring 2026

- Analyzed public perception and regulatory trends surrounding nuclear fusion integration into New Zealand's grid, synthesizing energy sector data to support strategic decision-making.
- Built interactive data visualization dashboards in Python and Streamlit to chart complex survey datasets, trendlines, and demographic distribution metrics.
- Authored technical summaries and outreach frameworks to effectively translate complex nuclear energy concepts into actionable recommendations for non-technical leadership and external stakeholders.